# Note 1.4 Matrix Algebra

## Algebraic Rules

- **Theorem 1.4.1:** Each of the following statements is valid for any scalars $\alpha$ and $\beta$ and any matrices $A, B, C$ .
  - $A + B = B + A$
  - $(A + B) + C = A + (B + C)$
  - $(AB)C = A(BC)$
  - $A(B + C) = AB + AC$
  - $(A + B)C = AC + BC$
  - $(\alpha\beta)A = \alpha(\beta A)$
  - $\alpha(AB) = (\alpha A)B = A(\alpha B)$
  - $(\alpha + \beta)A = \alpha A + \beta A$
  - $\alpha(A + B) = \alpha A + \alpha B$
- **Warning:** In general, $AB \neq BA$ .

## Notation

- **A $n \times n$ matrix multiplied by itself $k$ times ($k$ is a positive number):**

$$A^k = \underbrace{AA \cdots A}_{k \text{ times}}$$

## Application 1 A Simple Model for Marital Status Computations

- **Strategy:**
  Let a matrix $A$ whose first row is the percentages of married and single women who are married after 1 year, and the second row is the percentages of women who are single after 1 year. Then we let $\mathbf{x}$ be a vector whose entries are the number of married and single women. Then the number of married and single women after $n$ years can be determined by computing $A^n\mathbf{x}$ .
- **Example:**
  Simple model for marital status computations

## Application 2 Ecology: Demographics of the Loggerhead Sea Turtle

- **Strategy:**
  To create a Leslie matrix $L$ to correspond a population model and create a vector $\mathbf{x}_0$ to represent the initial population, then the population after $k$ years will be $L^k\mathbf{x}_0$.
- **Example:**
  Demographics of the Turtle

# The Identity Matrix

- **Definition:**
  The $n \times n$ identity matrix is the matrix $I = (\delta_{ij})$,where

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

- Remarks:
  - Application:
    For any $n \times n$ matrix $A$,

$$IA = AI = A$$

    For any $m \times n$ matrix $B$,

$$BI = B$$

    For any $n \times r$ matrix $C$,

$$IC = C$$

  - Standard notation:
    The column vectors of the $n \times n$ identity matrix $I$ are the standard vectors to define a coordinate system in Euclidean $n$-space. And it is denoted as $\mathbf{e}_j$ for the $j$th column vector of $I$. Thus the matrix $I$ can be written as

$$I = (\mathbf{e}_1, \mathbf{e}_2, \cdots, \mathbf{e}_3)$$

- **Zero Matrix:**
  Zero matrix is the matrix whose entries are all zero. (Denoted by $O$)

# Matrix Inversion

- **Definition:**
  An $n \times n$ matrix $A$ is said be nonsingular or invertible if there exists a matrix $B$ such that $AB = BA = I$. The matrix $B$ is said to be a multiplicative inverse of $A$.
  - singular:
    An $n \times n$ matrix is said to be singular if it does not have multiplicative inverse.
    When we can find a nonzero vector $\mathbf{x}$ with $A\mathbf{x} = \mathbf{0}$ or say we can't find the linear

combination of the column of $A$ to get $I$, then the matrix $A$ is singular since suppose there exists a matrix $B$ that is the multiplication inverse of $A$, then multiple both sides of the equation by $B$, then we get $\mathbf{x} = \mathbf{0}$, which is conflict with the proposition.

- Features:
  - A matrix can have at most one multiplicative inverse and the inverse of $A$ is denoted by $A^{-1}$.
  - Only square matrices have multiplicative inverses and can be said singular or nonsingular.
  - Any products of nonsingular is nonsingular.
- **Theorem 1.4.2:**
  If $A$ and $B$ are nonsingular $n \times n$ matrices, then $AB$ is also nonsingular and $(AB)^{-1} = B^{-1}A^{-1}$.
  - Remarks:
    If $A_1, \cdots, A_n$ are all nonsingular $m \times m$ matrices, then the product is nonsingular and

$$(A_1A_2\cdots A_n)^{-1} = A_n^{-1}\cdots A_2^{-1}A_1^{-1}$$

# Algebraic Rules for [[Note 1.3 Matrix Arithmetic#The Transpose of a Matrix|Transposes]]

- Rules:
  - $(A^T)^T = A$
  - $(\alpha A)^T = \alpha A^T$
  - $(A+B)^T = A^T + B^T$
  - $(AB)^T = B^T A^T$

# Application 3 Networks and Graphs

- Graph Theory:
  A technique using symmetric matrices to solve the problems involving networks.
- Basic concepts:
  - Graph: A set of points called vertices.
  - Edge: A set of unordered pairs of vertices. It can be represented by the vertices like $\{V_i, V_j\}$.
- **Representation for the network in matrices:**
  - If the graph contains a total of $n$ vertices, we can define an $n \times n$ matrix $A$ by

$$a_{ij} = \begin{cases} 1 & \text{if } \{V_i, V_j\} \text{ is an edge of the graph} \\ 0 & \text{if there is no edge joining } V_i \text{ and } V_j \end{cases}$$

    and the matrix is called the adjacency matrix of the graph. And all adjacency matrix are symmetric since if $\{V_i, V_j\}$ is an edge of the graph and then $a_{ij} = a_{ji} = 1$ and $a_{ij} = a_{ji} = 0$ if there is no edge joining $V_i$ and $V_j$. In either case, $a_{ij} = a_{ji}$.

- **Theorem 1.4.3:**
  - walk: a sequence of edges linking one vertex to another.
  - Theorem:
    If $A$ is an $n \times n$ adjacency matrix of a graph and $a_{ij}^{(k)}$ represents the $(i, j)$ entry of $A^k$, then $a_{ij}^{(k)}$ is equal to the number of walks of length $k$ from $V_i$ to $V_j$.

# Section 1.4 Exercises

- Question 1st
  - $(A+B)^2 = (A+B)(A+B) = (A+B)A + (A+B)B = A^2 + BA + AB + B^2$. In general, $BA + AB \neq 2AB$.
  - $(A+B)(A-B) = A^2 + BA - AB - B^2$. In general, $AB \neq BA$.
- Question 2nd
  If we take $a$ as an $n \times n$ matrix $A$ and $b$ as the $n \times n$ identity matrix $I$, then the rules in Question 1st work since $AI = IA$.
- Question 3rd
  Since $AB = O$, then we know that $\sum_{k=1}^{2} a_{ik} b_{kj} = 0$, that is, $a_{i1}b_{1j} + a_{i2}b_{2j} = 0$. Then, $\frac{a_{i1}}{a_{i2}} = -\frac{b_{2j}}{b_{1j}}$. So we can let

$$A = \begin{pmatrix} ka & a \\ tb & b \end{pmatrix} \quad B = \begin{pmatrix} -c & -d \\ kc & td \end{pmatrix}$$

- Question 12th
  Idea: Prove by definition that $AA^{-1} = A^{-1}A = I$. Note that we need to compute both $AA^{-1}$ and $A^{-1}A$.
- Question 22nd
  Definition:
  An $n \times n$ matrix $A$ is said to be **involution** if $A^2 = I$.
- Question 24th
  Definition:
  A matrix $A$ is said to be **idempotent** if $A^2 = A$.